DocQuorum

Agentic document extraction with open models:
the pipeline on real benchmark documents

October 2026

Every example is a real benchmark document and a stored pipeline output; no model was called to make these slides.

Outline

1. Task and pipeline
2. Walkthrough on real documents
   inputs, routing, parsing, readers, deterministic checks, selection by quorum, output
3. Results on six benchmarks; zoom re-reading
4. Summary

Setting

Open-weight models only: Qwen3.8-27B (FP8) for reading and MinerU2.5 for parsing scans, served on a single GPU. Each benchmark is scored with its own official scorer.

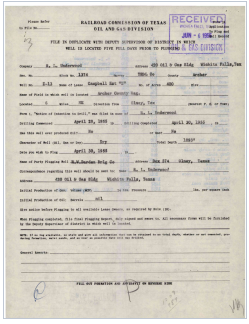
Problem statement

Task

Input: a document D (PDF or image), a JSON Schema S or a JSON template, and optionally a task description T.

Output: one JSON object Y that conforms to S, every value supported by D, absent values `null`.

Optional task (real example, receipt schema):
"A single point-of-sale purchase RECEIPT (retail or restaurant). DOC-TYPE / SCOPE RULE: if one file bundles MULTIPLE . . . "


RAILROAD COMMISSION OF TEXAS
OIL AND GAS DIVISION

RECEIVED

FILE IN DUPLICATE WITH DISTRICT SUPERVISOR OF DISTRICT IN WHICH WELL IS LOCATED FIVE DAYS PRIOR TO PLUGGING

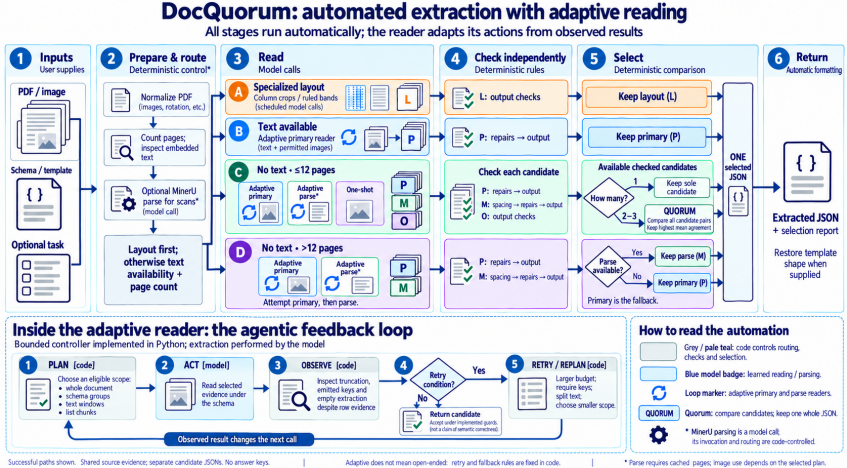
Scanned 1955 well-plugging application (Railroad Commission of Texas), 2 pages, no text layer.

JSON Schema (77 fields), excerpt

```
"p1_company":  {
  "anyOf":  [{"type":  "string"},
             {"type":  "null"}],
  "description":  "The 'Company' field
     in the identification block on
     page 1 ...  Extract verbatim as
     printed; null if blank.  Do NOT
     confuse with ..."
},
"p1_county":  {...},
"p1_total_depth":  {...},
...  74 further fields
```

ExtractBench short: 2A-9-160298_109929 (schema shortened); task text from the schema of hingham_lowes_receipt.

Pipeline overview



Pipeline figure (generated with Codex), describing the single-document entry point `docquorum.extract`; the lower panel shows the reader agent's plan–act–observe loop.

Step 2: the available evidence decides the route



A Layout reader

Unclaimed-property register
59 pages, 15,280 char./page



B Reader agent: text + images

County purchase order
1 page, 2,253 char./page



B Reader agent: text + images

W-14 well form
1 page, 252 char./page (OCR layer)



C Three readings, then quorum

1955 plugging application
2 pages, no embedded text



D Reader agent on the parse

Tax return
25 pages, no embedded text

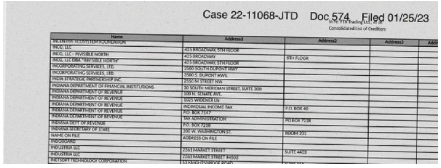
Routing rule

Layout first: pages printed as equal-width columns of records, or a ruled table on a scan, go to a layout reader A. Otherwise a PDF with embedded text (at least 50 characters per page; a digital PDF or an OCR layer) is read once B. A scan without text is parsed by MinerU and read three ways if it has at most 12 pages C, or read through its parse D.

ExtractBench documents; embedded characters per page measured with pdftotext.

Step 2: MinerU returns a scanned table as cells

Scanned page (detail)



MinerU output (raw HTML, abbreviated)

```
<tr><td>Name</td><td>Address1</td><td>Address2...<td>INCENTIVE
ECOSYSTEM FOUNDATION</td><td...<td>INCO, LLC</td><td>425
BROADWAY, 5TH FL...
```

The same rows as table cells (65 rows on this page)

Name	Address1	City	State	PostalCode
INCENTIVE ECOSYSTEM FOUNDATION				
INCO, LLC	425 BROADWAY, 5TH FLOOR			
INO, LLC - INAVAILABLE NORTH	425 BROADWAY	NEW YORK	NY	10013
INO, LLC DBA "INAVAILABLE NORTH"	425 BROADWAY, 5TH FLOOR	NEW YORK	NY	10013-2599
INCORPORATING SERVICES, LTD.	3500 SOUTH COOPONT HWY	NEW YORK	NY	10013

Columns stay aligned: an empty cell stays empty rather than pulling the next value into its place. Recognition errors remain, so the reader receives the page image together with the parse.

ExtractBench long: real_ftx_full_corrupted, page 47; cached MinerU2.5 parse.

Step 3, route A: the column reader



COLLINS RICHARD
EATON BETHINA
RT 1 BOX 220
JOHNSON JOHNEY B
C/O MAE JEAN JOHNSON,
RT 1 BOX 230
SHERRILL IVANNA
RT 1 BOX 1142

PROCTOR

BLEVINS JAMES DEE
RR2 BOX 556
SANDRA COTTONGAME
67000 S 367 RD

STILWELL

ADAIR TERI
RT 2 BOX 1155
BEASLEY ALAN C
470241 E 880 RD
BRD ALFRED JOSEPH
RT 2 BOX 271A
BREWER VENA MAUD
301 S CHEROKEE ST
BRIDGES INEZ
77150 HIGHWAY 59
BROWN CLIFFORD LEE
RT 4 BOX 2570
BROWN LISA
RT 3 BOX 2119
CRAWFORD BRADLEY
212 6TH PL
CRITTENDEN JESSIE
RT 6 BOX 1760
EUBANKS JR SHELBY
RT 4 BOX 1474
FOURKILLER HENRY
471958 E 810 RD

Records returned for column 1 (page 2)

name	locale
COLLINS RICHARD	null
EATON BETHINA	RT 1 BOX 220
JOHNSON JOHNEY B	C/O MAE JEAN JOHNSON, RT 1 BOX 230
SHERRILL IVANNA	RT 1 BOX 1142

7
columns detected on the page

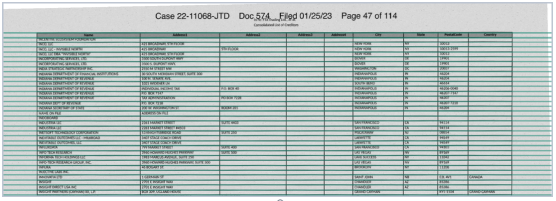
59
pages

26,675
records in the final output

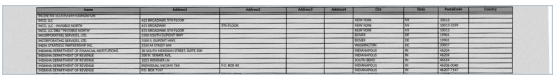
The trigger is page geometry only: at least three text columns of near-equal width. Each column is cut out and read top to bottom in one call; a record continued from the previous column is joined to it.

ExtractBench long: real_oklahoma_unclaimed_2024, page 2. Column boxes drawn with the pipeline's own geometry code; records from the stored final output.

Step 3, route A: the ruled-table reader



Printed rules detected after deskewing by 1.15° (green).



One call: the header row stacked on a band of 12 rows; the reply must contain exactly 12 entries.

ExtractBench long: real_ftx_full_corrupted, page 47. Rows from the stored final output.

67

rows counted from the rules

67

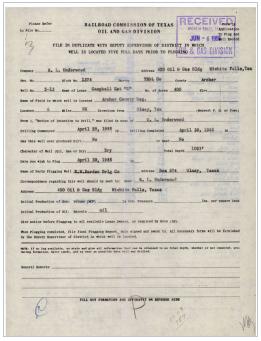
rows returned by the reader

Bands on page 47: $12 + 12 + 12 + 12 + 12 + 7$.

name	address_1	city	state
INCENTIVE ECOSYSTEM FOUNDATION	null	null	null
INCO, LLC	425 BROADWAY, 5TH FLOOR	NEW YORK	NY
INCO, LLC - INVISIBLE NORTH	425 BROADWAY	NEW YORK	NY

Fixing the number of entries per band prevents repeated identical rows from multiplying or disappearing.

Step 3, route C: three independent readings of one scan



field	P image reading	M parse reading	O one-shot
p1_company	R. L. Underwood	R. L. Underwood	null
p1_county	Archer	Archer	null
p1_well_no	Z-13	Z-13	null
p1_name_of_lease	Campbell Est "Z"	Campbell Est "Z"	null
p1_drilling_commenced	April 29, 1955	April 29, 1955	null
p1_party_plugging_well	R.W. Darden Dr'lg Co	R.W.Darden Drlg Co	null
p2_correspondence_name	R.L. Underwood	R.L.Underwood	null

Red: empty, or the two readings disagree (P and M differ only in spacing and punctuation).

P 59 / 77 fields filled

M 59 / 77 fields filled

O 1 / 77 fields filled

P: the reader agent on the page images. M: the reader agent on the MinerU parse and the images. O: the benchmark's own one-shot reader.

ExtractBench short: 2A-9-160298_109929; stored readings after their checks.

Step 4: checks that repair the structure of a reading

Check	Document	Observed change	Value F1
Rows copied into another list	Pueblo check register (medium)	**562 → 51 batch rows** Check 01033982 ($700.88) had also been filed as a batch with the same number and amount.	84.73 → 99.91
Identical records capped at the printed count	FTX creditor list, text layer (long)	**146 → 37 rows on page 59** Page 59 prints the line "NAME ON FILE" exactly 37 times.	84.47 → 90.77
Stacked lines folded into one record	Clinical adverse-event listing (long)	**1,499 → 668 records** Subject R1002: three printed lines became one event with its resolved date.	38.80 → 68.77
Template rows dropped	Texas W-14 well form (short)	**5 → 2 casing rows** Three rows carried only the printed row label (e.g. "Intermediate").	82.30 → 93.23

Examples are the cases each check was designed for. On a few short documents some repairs lowered the score slightly (e.g. duplicate removal on three election reports, −3.5 points in total).

Stored readings before and after each check, re-run with `docquorum/checks` and scored with ExtractBench's scorer. Each check uses only the schema and the document's own text; no model is called.

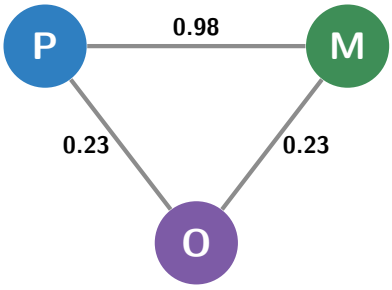
Step 4: checks that make values exactly what the page prints

Check	Document	Observed change	Value F1
Every schema field present	Caterpillar 330 spec sheet (short)	**missing keys → `null`** The reply omitted 7 optional fields in every specification row; the scorer counts an absent key as a miss.	61.48 → 95.74
A blank tick box is false	Texas P-4 form (short)	**6 boxes: `null` → `false`** The field description states "false if left blank".	91.67 → 96.67
Quotes as printed	Census statistical table (short)	**Bachelor's → Bachelor's** The page prints a typographic apostrophe; only the quote character is copied from the page.	60.12 → 62.58
Numbers as printed	Schedule K-1 (short, one-shot reading)	**33.333333 → 33.3333333** A rounded value is replaced by the single printed figure it restates (6 share fields).	85.42 → 97.92

Every replacement is a value that the document itself prints, or `null` where the schema admits null.

Stored readings before and after each check, re-run with `docquorum/checks` and scored with ExtractBench's scorer. Each check uses only the schema and the document's own text; no model is called.

Step 5: selection by quorum



Edge: agreement of two readings, measured with the benchmark's value scorer, recall-weighted (F_β, $\beta = 2$).

Reading	Mean agreement	
P image reading	0.604	**kept**
M parse reading	0.604	
O one-shot	0.218	

P and M tie; ties go to the reading listed first. The whole of P is returned; fields are never mixed across readings.

Agreement is not accuracy

Answer-key value F1 for this document (illustration only; never used by the pipeline): P 95.24, M 95.24, O 22.50. On the scanned county purchase order the quorum kept the one-shot reading (78.26), although the image reading scored higher (82.61).

ExtractBench short: 2A-9-160298_109929; agreement computed by `docquorum.quorum.pick` on the checked readings.

Step 6: one checked JSON, in the form requested

JSON Schema in, JSON out (1955 form, 77 fields)

```
{
  "p1_company":  "R. L. Underwood",
  "p1_address":  "420 Oil & Gas Bldg Wichita
Falls, Tex",
  "p1_county":  "Archer",
  "p1_well_no":  "Z-13",
  "p1_name_of_lease":  "Campbell Est \"Z\"",
  "p1_drilling_commenced":  "April 29, 1955",
  "p1_total_depth":  "1093'",
  ...  70 further fields
}
```

JSON template in, the same template filled



Template

```
{"menu":  [{"price":  "", "nm":  ""}],
 "total":  {"cashprice":  "",
            "total_price":  ""}}
```

Output (equal to the answer key)

```
{"menu":  [
   {"price":  "17500", "nm":  "J.STB PROMO"},
   {"price":  "46000", "nm":  "Y.B.BAT"},
   {"price":  "27500", "nm":  "Y.BASO PROM"},
 ],
 "total":  {"cashprice":  "91000",
            "total_price":  "91000"}}
```

Left: stored final output for 2A-9-160298_109929. Right: UNIKIE-BENCH Catering-Services test_1.jpg, quorum over six stored readings.

ExtractBench: complete results (1/3)

#	System	Type	Short	Medium	Long	Overall
1	LlamaExtract Agentic Plus	Commercial service	96.94	94.45	90.18	95.91
1	Pulse (Effort)	Commercial service	96.46	95.01	93.51	95.91
3	**DocQuorum (ours)**	**Open model (27B)**	**96.34**	**94.11**	**90.41**	**95.43**
4	LlamaExtract Agentic	Commercial service	96.04	92.11	91.83	94.77
5	Codex (GPT-6 Sol Evidence)	Coding agent	95.99	92.42	87.25	94.57
6	Codex (GPT-5.6 Sol Evidence)	Coding agent	96.04	90.19	82.73	93.77
7	Codex (GPT-5.5)	Coding agent	95.68	91.15	78.88	93.57
8	Codex (GPT-5.5 Evidence)	Coding agent	95.59	88.72	87.97	93.35
9	Pulse (Non-effort)	Commercial service	93.17	92.22	89.07	92.70
10	LlamaExtract Cost-Effective	Commercial service	94.03	89.51	83.73	92.27
11	Codex (GPT-5.6 Terra Evidence)	Coding agent	95.60	86.69	77.36	92.26
12	OpenAI GPT-6 Astra	Commercial model	97.22	90.56	31.70	91.91
13	Codex (GPT-5.6 Luna Evidence)	Coding agent	93.50	87.43	80.05	91.16
14	Codex (GPT-6 Luna Evidence)	Coding agent	93.09	87.71	80.88	91.01
15	Reducto Deep Extract	Commercial service	94.20	80.47	92.01	90.44
16	Qwen3.8 Flash Next	Open model	94.82	87.81	37.74	89.88

ExtractBench leaderboard, snapshot of 2 October 2026 (47 published systems) and DocQuorum. Value F1 (%) by the benchmark's scorer over 370 PDFs (252 short, 98 medium, 20 long); Overall is the mean over all documents. Equal scores share a rank.

ExtractBench: complete results (2/3)

#	System	Type	Short	Medium	Long	Overall
17	Qwen3.8 27B	Open model	94.68	87.54	38.45	89.75
18	Extend (Max Context)	Commercial service	92.32	78.75	90.46	88.62
19	Qwen3.6 35B	Open model	92.86	86.37	36.90	88.11
20	Qwen3.5 35B	Open model	93.52	85.27	31.73	88.00
21	DeepSeek V4.1 Flash	Commercial model	94.44	81.49	22.23	87.11
22	Claude Code (Opus 4.8)	Coding agent	90.08	79.21	88.07	87.09
23	Claude Code (Opus 4.8 Evidence)	Coding agent	90.54	79.17	77.94	86.84
24	Qwen3.5 9B	Open model	92.31	81.12	32.22	86.10
25	Claude Code (Opus 5.5 Evidence)	Coding agent	89.63	77.96	80.83	86.06
26	Extend Extract	Commercial service	89.39	84.14	47.21	85.72
27	Datalab (Accurate + Balanced)	Commercial service	89.40	85.08	42.04	85.70
28	Reducto Extract	Commercial service	92.97	69.59	30.65	83.41
29	Kimi K3	Commercial model	94.64	69.64	4.99	83.17
30	Qwen3.5 4B	Open model	89.06	76.64	27.31	82.43
31	NuExtract3	Open model	86.41	77.01	32.71	81.02
32	GLM-5.3 Flash	Commercial model	96.30	51.56	27.83	80.75

ExtractBench leaderboard, snapshot of 2 October 2026 (47 published systems) and DocQuorum. Value F1 (%) by the benchmark's scorer over 370 PDFs (252 short, 98 medium, 20 long); Overall is the mean over all documents. Equal scores share a rank.

ExtractBench: complete results (3/3)

#	System	Type	Short	Medium	Long	Overall
33	Google Gemini 3.8 Flash	Commercial model	88.14	72.23	28.70	80.71
34	Google Gemini 3.5 Flash	Commercial model	87.87	69.76	27.90	79.84
35	Lift Datalab 9B	Open model	87.64	65.69	24.78	78.43
36	OpenAI GPT-5.4 Nano	Commercial model	77.43	76.37	35.81	74.90
37	Gemma4 26B	Open model	77.70	59.61	28.88	70.27
38	Gemma4 E4B	Open model	81.10	48.93	26.82	69.64
39	InternVL3.5 14B	Open model	80.70	39.04	19.41	66.35
40	InternVL3.5 8B	Open model	79.26	36.62	17.57	64.63
41	Qwen3.5 2B	Open model	71.22	50.93	25.93	63.40
42	Kimi-VL A3B	Open model	69.86	34.61	18.94	57.77
43	Gemma4 E2B	Open model	63.71	28.96	14.58	51.85
44	NuExtract 2.0 8B	Open model	62.92	23.50	6.39	49.42
45	Qwen3.5 0.8B	Open model	57.44	26.98	6.65	46.63
46	MiniCPM-V 4.5	Open model	57.39	25.30	1.13	45.85
47	Gemma4 12B	Open model	53.62	30.82	13.16	45.39
48	GLM-4.6V Flash	Open model	47.28	44.52	22.17	45.19

ExtractBench leaderboard, snapshot of 2 October 2026 (47 published systems) and DocQuorum. Value F1 (%) by the benchmark's scorer over 370 PDFs (252 short, 98 medium, 20 long); Overall is the mean over all documents. Equal scores share a rank.

UNIKIE-BENCH: complete results

#	System		Ret.	Cat.	Com.	Acc.	Post.	Adm.	Edu.	Adv.	Tax	Med.	Nut.	Avg.
1	**DocQuorum (ours)**	O	**90.78**	**88.19**	**53.46**	**80.00**	**85.65**	**86.95**	**85.46**	**86.21**	**82.13**	**77.71**	**92.26**	**82.62**
2	Gemini-3-Pro	C	89.27	83.54	53.87	81.43	90.74	84.32	80.71	90.54	81.54	77.60	92.47	82.37
3	Qwen3-VL-Plus	C	89.12	86.45	50.57	77.86	83.55	78.91	87.44	83.14	78.93	76.46	89.72	80.20
4	Qwen3-VL-8B	O	88.26	86.23	51.14	71.07	83.80	78.90	87.39	81.23	78.74	77.19	86.36	79.12
5	Qwen-VL-Max	C	86.67	84.68	49.33	76.07	84.85	79.96	82.46	82.76	77.94	75.62	75.14	77.77
6	GLM-4.1V-9B	O	62.32	77.33	46.67	76.43	76.46	78.75	65.83	81.99	70.68	72.08	60.47	69.91
7	GPT-4o	C	75.79	79.33	49.67	67.86	82.95	64.26	47.77	73.56	75.33	67.57	76.51	69.15
8	MiMo-VL-7B-RL	O	77.45	77.66	49.01	67.50	76.70	76.81	76.65	66.28	74.86	68.70	46.10	68.88
9	InternVL3.5-8B	O	77.23	72.23	41.64	67.14	73.90	72.10	63.08	73.08	72.81	65.94	65.53	67.70
10	GPT-5	C	71.90	71.33	45.94	69.29	80.95	62.79	50.26	75.48	72.05	67.71	75.83	67.59
11	MiniCPM-V4.5-8B	O	68.37	74.91	45.91	67.76	83.60	71.89	63.21	79.31	69.57	70.00	44.18	67.15
12	Claude-Sonnet-4.5	C	69.24	73.65	45.92	69.29	79.45	69.20	39.46	85.06	67.79	63.96	67.24	66.39
13	Kimi-VL-A3B	O	86.49	63.12	42.87	61.79	83.40	68.72	80.37	72.41	77.80	64.93	25.72	66.14
14	Gemma-3-12B	O	54.32	68.73	30.68	53.93	72.30	37.14	17.56	36.78	56.81	48.54	52.20	48.09
15	Ministral-3-8B	O	6.47	57.01	26.24	43.39	75.80	32.37	29.54	19.53	65.85	51.46	6.17	37.62
16	SmolVLM2-2.2B	O	17.80	57.22	28.52	18.92	34.98	8.63	3.87	27.16	34.97	18.00	10.51	23.69

Ret. Retail; Cat. Catering services; Com. Commercial; Acc. Accommodation; Post. Postal label; Adm. Administrative; Edu. Education; Adv. Advertisement; Tax: Tax-compliant; Med. Medical services; Nut. Nutrition label; Avg. macro average. C: closed model; O: open model. Underlined: best in the column.

UNIKIE-BENCH (ACL 2026), Table 4: 15 published systems; F1 (%) by the benchmark's evaluator on 4,472 images, every input capped at 1,605,632 pixels. DocQuorum: quorum over six readings from three open models; small images enlarged before reading.

Results on six benchmarks

Benchmark	DocQuorum	Place	Comparison
VAREX (1,777 government forms) (with zoom re-reading)	**98.30**	**1st of 21**	next: Gemini 2.5 Pro 98.0
UNIKIE-BENCH (4,472 images)	**82.62**	**1st of 16**	next: Gemini-3-Pro 82.37
LongArray-Extract (45 long documents)	**98.05**	2nd of 12	1st: Extend MAX 99.2
UNIKIE-BENCH open category (1,661 documents) (with zoom re-reading)	**72.33**	2nd of 16	1st: Gemini-3-Pro 81.65
ExtractBench (370 PDFs) (development set)	**95.43**	3rd of 48	1st: LlamaExtract Agentic Plus 95.91
DocuBench (72 documents, 10 file types)	**86.47**	5th of 10	1st: DocuPipe (high effort) 97.02

Each score is the benchmark's own scorer on the full test set, with open-weight models (Qwen3.8-27B and MinerU2.5) on one GPU. Places count every published system.

docs/results/: one complete table per benchmark (TABLES.md). No result was submitted to a leaderboard.

Zoom re-reading: a new step after the quorum

After the quorum keeps one reading, each field on which the readings disagree is found on the page and read once more from a sharp crop of that region.

This new reading sees none of the earlier answers. The kept value changes only if it matches a value some reading already proposed.

The design was fixed on ExtractBench, then run once on two other benchmarks.

Held-out results

VAREX: 98.10 → **98.30**
60 forms better, 25 worse

UNIKIE open: 71.07 → **72.33**
291 documents better, 138 worse

Both are far beyond chance (sign test).

results/README.md: method, statistics, and every changed value checked against the answer key after the runs.

Summary

Results

1st on VAREX and UNIKIE-BENCH; 2nd on LongArray-Extract and UNIKIE open; 3rd on ExtractBench.

A new step, zoom re-reading, improved both benchmarks it was tested on.

Open-weight models on one GPU.

Method

Routing by the available evidence; an agentic reader that plans, reads and re-reads; layout readers for column flow and ruled tables; deterministic checks against the schema and the page; selection by quorum.

Limitations

ExtractBench served as the development set. Agreement among readings does not guarantee accuracy. Checks are triggered by schema wording and page text; a few repairs lower the score on some documents.